WEEK 2 PROJECT

Python & Pandas — 20 Practical Exercises

Beginner-friendly guide for your Data Scientist internship

Portal task: Complete 20 Pandas exercises on filtering, grouping, and merging. The screenshot shows a due date of 29 October 2026.

What to submit: A tested Python notebook containing the 20 exercises and a short note explaining what you built and learned.

Easy setup: Open https://colab.research.google.com/, sign in, choose New notebook, and rename it Week2_Python_Pandas.ipynb. This avoids installing software.

1. Create the practice dataset

```
import pandas as pd

students = pd.DataFrame({
    "student_id": [1, 2, 3, 4, 5, 6, 7, 8],
    "name": ["Anu", "Ravi", "Meena", "Kiran", "Sita", "Arjun", "Divya", "Rahul"],
    "department": ["AI&DS", "CSE", "AI&DS", "ECE", "CSE", "AI&DS", "ECE", "CSE"],
    "city": ["Kadapa", "Hyderabad", "Kadapa", "Tirupati", "Hyderabad", "Vijayawada", "Kadapa", "Tirupati"],
    "attendance": [92, 78, 88, 65, 95, 81, 73, 86],
    "marks": [88, 67, 91, 55, 94, 76, 62, 84],
    "fees_paid": [True, True, False, True, True, False, True, True]
})
print(students.head())
print(students.info())
```

Run this cell first. It creates sample data in memory; no file download is needed.

2. Complete these 20 exercises

1. Show the first five rows

```
students.head()
```

2. Show the last three rows

```
students.tail(3)
```

3. Check dataset dimensions

```
students.shape
```

4. Display column names

```
students.columns
```

5. Check data types and missing values

```
students.info()
```

6. Select name and marks columns

```
students[["name", "marks"]]
```

7. Filter marks above 80

```
students[students["marks"] > 80]
```

8. Filter attendance of 80 or more

```
students[students["attendance"] >= 80]
```

9. Filter AI&DS; students

```
students[students["department"] == "AI&DS"]
```

10. Find students who paid fees

```
students[students["fees_paid"] == True]
```

11. Sort marks highest to lowest

```
students.sort_values("marks", ascending=False)
```

12. Sort attendance lowest to highest

```
students.sort_values("attendance")
```

13. Count students in each department

```
students.groupby("department")["student_id"].count()
```

14. Average marks by department

```
students.groupby("department")["marks"].mean()
```

15. Average attendance by department

```
students.groupby("department")["attendance"].mean()
```

16. Highest marks by department

```
students.groupby("department")["marks"].max()
```

17. Multiple summaries by department

```
students.groupby("department").agg(student_count=("student_id","count"), average_marks=("marks","mean"), highest_marks=("marks","
```

18. Add Pass/Fail result column

```
students["result"] = students["marks"].apply(lambda x: "Pass" if x >= 40 else "Fail")
students[["name", "marks", "result"]]
```

19. Merge students with department mentors

```
mentors = pd.DataFrame({"department":["AI&DS","CSE","ECE"], "mentor":["Ms. Rao","Mr. Kumar","Ms. Fatima"]})
students.merge(mentors, on="department", how="left")
```

20. Merge a fee table with student data

```
fee_details = pd.DataFrame({"student_id":[1,2,3,4,5,6,7,8], "fee_amount":[25000,25000,25000,22000,25000,25000,22000,25000]})
final_df = students.merge(fee_details, on="student_id", how="left")
final_df[["name","department","fees_paid","fee_amount","marks"]]
```

Important: Run exercises 1–17 in order. Exercise 18 creates a result column; exercises 19–20 create extra DataFrames and merge them. Put each exercise in a separate notebook cell and check the output.

3. Understand the main concepts

Filtering: Selects rows that match a condition, for example marks above 80.

Grouping: groupby() creates groups such as departments so you can calculate counts, averages, or maximum values.

Merging: merge() combines DataFrames using a shared column such as department or student_id. A left merge keeps all rows from the left DataFrame.

Sorting: sort_values() arranges records in ascending or descending order.

4. Project description for submission

Title: Python and Pandas — Data Filtering, Grouping, and Merging

This project practises data manipulation using Python and the Pandas library. A sample student dataset is created with student IDs, names, departments, cities, attendance, marks, and fee status. Twenty exercises are completed to inspect data, select columns, filter records, sort values, group students by department, calculate summary statistics, create a result column, and merge related DataFrames. The project demonstrates how Pandas can organise and summarise structured data efficiently. It strengthens basic data-cleaning, analysis, and problem-solving skills useful in data analytics.

Learning outcomes: I practised DataFrame inspection, conditional filtering, sorting, groupby(), aggregate functions, apply(), and merge(). I learned how to summarise structured data and combine information from multiple tables.

Make it accurate: Submit only after running the exercises. Replace the sample data if your mentor gave you a required dataset.

5. Final submission checklist

• Save the notebook as Week2_Python_Pandas.ipynb.

• Run the dataset cell and all 20 exercises.

• Check outputs and fix errors.

• Take screenshots of selected outputs if requested.

• Submit the notebook and description using the portal's Submit work button.

• Confirm the portal shows a submitted status.

Time plan: Setup (15 minutes), exercises 1–10 (30 minutes), exercises 11–20 (40 minutes), testing and submission (15 minutes). Split the work into short sessions around your semester exams.